

Assignment #5: Database Design

Description:

The goal is to practice database design using normalization.

Work the Morgan Importing Problem described as described below. No database development is required for this assignment, just the table design. The assignment involves concepts covered in chapter 4 and its slides. Pay attention to the following guidelines for Assessing Table Structure from Figure 4-1 of Chapter 4.

- Count rows and examine columns
- Examine data values and interview users to determine:
 - Multivalued dependencies
 - Functional dependencies
 - Candidate keys
 - Primary keys
 - Foreign keys
- Assess validity of assumed referential integrity constraints

Assessing Table Structure

Morgan Importing Problem:

Phillip Morgan, the owner of Morgan Importing, makes periodic buying trips to various countries. During the trips, he keeps notes about the items he purchases and basic data about their shipments. He hired a college student as an intern, and she transformed his notes into the spreadsheets in Figure 4-9. This is just sample data. Phillip has purchased hundreds of items over the years, and they have been shipped in dozens of different shipments.

Phillip wants to enter the information age, thus he has decided to develop a database of his inventory. He wants to keep track of the items he has purchased, their shipments, and eventually customers and sales. To get started, he has asked you to create a database for the data in Figure 4-9.

	A	B	C	D	E	F	G	H	I
1	ShipmentNumber	Shipper	Phone	Contact	From	Departure	Arrival	Contents	InsuredValue
2	49100300	Wordwide	800-123-4567	Jose	Philippines	5/5/2009	6/17/1999	QE dining set, large bureau, porcelain lamps	\$27,500
3	488955	Intenational	800-123-8898	Marilyn	Singapore	6/2/2009		Miscellaneous linen, large masks, 14 setting Willow design china	\$7,500
4	84899440	Wordwide	800-123-4567	Jose	Peru	7/3/2009	7/28/2009	Woven goods, antique leather chairs	
5	399400	Intenational	800-123-8898	Marilyn	Singaporeee	8/5/2009	9/11/2009	Large bureau, brass lamps, willow design serving dishes	\$18,000
6									
7									
8									
9		Item	Date	City	Store	Salesperson	Price		
10		Willow Serving Dishes	7/15/2009	Singapore	Jade Antiques	Swhee Lai	\$4,500		
11		Large bureau	7/17/2009	Singapore	Eastern Sales	Jeremey	\$9,500		
12		Brass lamps	7/20/2009	Singapore	Jade Antiques	Mr. James	\$1,200		
13		QE Dining Set	4/7/2009	Manila	E. Treasures	Gracielle	\$14,300		
14									
15									

Figure 4-9 – Spreadsheets from Morgan Importing

To complete this assignment, you must complete following instructions and provide detailed answers for each item:

- A *Follow the “Assessing Table Structure” procedure shown. List multivalued dependencies, functional dependencies, candidate keys, primary keys, and foreign keys. State your assumptions.*
 - B *List questions you would ask Phillip to verify your assumptions.*
 - C *Create tables as necessary to eliminate multivalued dependencies, if any.*
 - D *The relationship between shipment and item data could be inferred by matching values in the From cells to values in the City cells. Describe two problems with that strategy.*
 - E *Describe a change to this spreadsheet that does express the shipment/item relationship.*
 - F *Assume that Phillip wishes to create an updateable database from this data. Design tables you think are appropriate. State all referential integrity constraints.*
 - G *Assume that Phillip wishes to create a read-only database from this data. Design tables you think are appropriate. State all referential integrity constraints.*
 - H *Do these data have the multivalued, multicolumn problem? If so, how will you deal with it?*
 - I *Do these data have the inconsistent data problem? If so, how will you deal with it?*
 - J *Do these data have a null value data problem? If so, how will you deal with it?*
 - K *Do these data have the general-purpose remarks problem? If so, how will you deal with it?*
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For part A, make sure you state your assumptions.

For part B, list at least 6 questions to verify your assumptions stated in part A.

For part C, describe the tables using the textbook format (the PK is underlined and the FK is italicized). For example, **SKU_DATA** (**SKU**, *SKU_Description*, *Department*, *Buyer*)

For part D, focus on the FK usage and the issue of data consistency when addressing this part.

For Parts F and G, , describe the tables using the textbook format (underlined PK and italicized FK) with referential integrity statements. For example,

SKU_DATA (**SKU**, *SKU_Description*, *Department*, *Buyer*)
BUYER (**BuyerName**, *Department*)
Where SKU_DATA.Buyer must exist in BUYER.BuyerName

Include all of your answers for all parts in one MS Word document. See other useful hints document in module 5.